Paper trading the Day Trader skill

13 scenarios on the same 39 recorded TSLA sessions, 24 Aug 2026 to 16 Oct 2026

Run on 3 November 2026 with the repo's own tools and data. $5,000 practice account. Every number here comes from the files in `paper-trading-scenarios/`.

Each scenario is the README example's rules with **one choice changed**: the stop, the target, the entry order, the opening range, a volume check, the setup, the risk per trade, the entry window, or the Time Machine as a filter. Nine choices give twelve alternative settings, plus the baseline, so any difference traces back to that one choice. For every session, the pack's own tools found the signal, planned the exit, replayed it in the Time Machine, sized it, ran the risk check and showed the ticket. Where the ticket was READY, it got an automatic practice yes, a simulated paper broker filled the exact order the ticket showed against that session's 5-minute bars, and the tools recorded every placement and fill, as on a real day.

What happened

- **The pipeline ran on all 507 scenario sessions** (13 × 39) without a tool error: 220 trades, the rest no signal, plus 2 risk blocks and 2 filter skips. Every READY ticket got an automatic practice yes (`ticket.py approve --auto-yes`), and nothing was sent to a broker. The tools' own books matched the simulated broker's cash every evening.
- **2 of 13 ended above $5,000:** S10 Setup: prior-day high (+$59.42) and S04 Stop: 1 % (+$50.74). The baseline ended at -$62.52 over 19 trades (6 targets, 9 stops, 4 sold at 15:55). The worst was S06 Target: 2R at -$119.50.
- **None of it is beyond chance.** Every scenario with ten or more trades has a 90 % bootstrap range that runs from a loss to a gain, so none can be told apart from break-even, or from the others. With 17 to 19 trades, one trade ending at the target instead of the stop moves a total by about $67.04.
- **Maximum loss at once.** In the twelve $50 scenarios, no open position was ever down more than -$49.60 (S03 on 17 Sep at 11:00). With the $100 limit the worst moment was -$62.01. No stop was gapped through. Once, a market fill above the plan's price took a position's risk past the limit ($55.36 at the stop, S02 27 Aug); it never hit the stop. Over several sessions the losses added up: drawdowns of $42.71 to $210.70, which the daily limit ($150, one day at a time) never sees.
- **The runs turned up three things in the tools** (see the last pages): size.py and check.py measure the position cap against different bases, so two S11 orders were sized and then blocked. Market fills above the plan's price carried up to $7.40 more risk than the risk check approved. An entry watch that ends without a trigger leaves the previous day's `entry.json` in place (size.py then refuses the old plan, so no order comes of it).

Read this first

Not financial advice. This is one stock over eight weeks, replayed with simulated fills. It shows what these rules did on these days, not what they will do on the next ones. Fills come from 5-minute bars without fees, spread or partial fills. No broker was connected.

The setup, and how each session ran

Setup done on this PC (howtouse.md, section 1)

Step	What was done
python3	Copied `python.exe` to `python3.exe` in the Python 3.13 folder, so the skills' `python3` commands run the real Python (checked in Git Bash: Python 3.13.5).
Skills	Copied the 11 `trader-*` folders into `%USERPROFILE%\.claude\skills`, so `/trader-scanner` and the rest appear in new Claude Code sessions.
Workspace	Made `%USERPROFILE%\trading-desk\trader` with `config.json`, a copy of the example rules ($5,000, $50 a trade, $150 a day, paper).
TRADER_HOME	Set to that folder (user environment variable), so every tool uses it from any folder. First fixed the tests so they ignore it: before the fix, setting it made 12 tests fail and wrote test files into that folder.
Checked	All 384 tests pass, with and without TRADER_HOME. The offline replay of the README example gives the README's numbers.
Left for you	Your name and email in `sec_user_agent` (the SEC tools send it). The Alpaca paper connection (howtouse.md section 4) needs your own keys and dashboard. Add `alpaca_account` only after you have read the number in the Alpaca dashboard yourself.

One session, in one scenario

1 Entry
entry.py watch replays the session bar by bar from 9:30 and stops at the trigger.

2 Exit plan
exit.py plan sets the stop and target before any order.

3 Time Machine
10,000 replays of the rest of the day, plus every past session of the rule.

4 Size
size.py fits whole shares inside the four limits and writes one bracket order.

5 Risk
check.py gives PASS or BLOCK and records the verdict.

6 Ticket and yes
ticket.py show turns a PASS into a READY ticket; approve --auto-yes gives the yes.

7 Paper broker
Simulated: fills that order on the session's bars; the tools record it.

Steps 1 to 6 are the real tools, run from the repo on a **pinned clock** (practice runs), in the scenario's own trader folder. Each morning the folder is cleared of the previous day's plan files, and the account starts with the cash the tools recorded the evening before. **The yes in step 6 is automatic**: `ticket.py approve --auto-yes`, which the tool accepts only on a practice run with `"paper": true`, so nothing can be placed from it; on the real clock it refuses, and a real order still needs your typed yes. Step 7 stands in for the Allow once card and Alpaca paper. The tools then record the placement and every fill (`ticket.py result`, `fill --practice`), and a position still open at 15:55 goes out through `flatten`, the risk check and its own automatic yes, so the risk check counts each trade the way it would on a real day.

The simulated paper broker

- **Market entry:** fills at the open of the bar after the signal, the first price after the trigger bar closed.
- **Limit entry:** fills at that open if it is at or under the limit, else at the limit once a bar trades down to it. If it has not filled by the end of the entry window, it is canceled.
- **Stop-loss:** sells at the stop, or at the bar's open if a bar opens below the stop (a gap). **Take-profit:** sells at the target, or at the open if a bar opens above it.
- **Both in one bar:** the stop is assumed to come first, because 5-minute bars do not say which came first. **Flat by:** anything still open at 15:55 sells at that bar's open, once the bracket's two legs are canceled; the sell is the flatten ticket the tools wrote, checked and approved.
- **Open loss** is marked at every bar's low (or the stop fill), so the deepest open loss is the most the position was down at any one moment.

The data is the repo's 60 days of TSLA 5-minute bars (`tests/fixtures/trader-timemachine/TSLA-60d5m.json.gz`, 59 full sessions). The first 20 sessions serve as history only, because the Time Machine needs 20 sessions before the day it replays. No network was used. A separate check recomputed all 220 trades from the raw bars and got the same fills, exits and results.

The 13 scenarios

The baseline is the README example's trade rule, run every session. Each other scenario changes one thing and keeps the rest: the 15-minute opening range on 5-minute bars, a close above the range high, the stop at the range midpoint, the target at 1R, a market entry, and the example config: $5,000 budget, $50 max loss a trade, $150 a day, 3 trades a day, a 75 % position cap, a 10 % cash buffer, minimum reward:risk 1.0, entries 09:45 to 15:30, flat by 15:55.

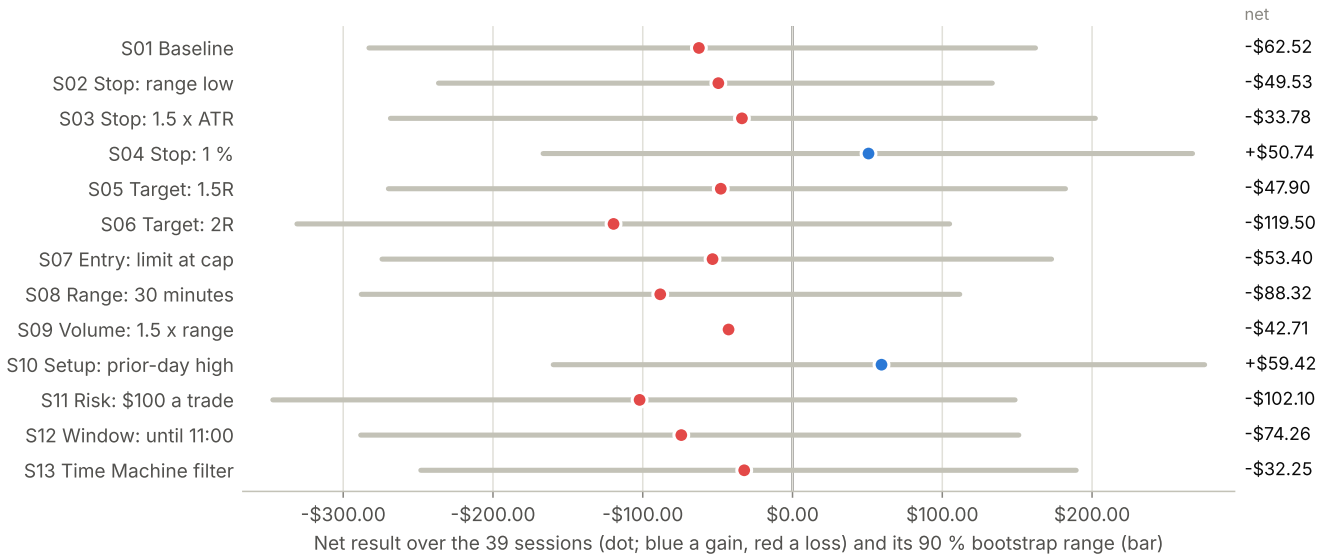
#	Feature	This scenario	The baseline has
S01	Baseline	The README's rules: 15-minute range, stop at the range midpoint, 1R target, market entry, $50 a trade	
S02	Stop rule	Stop at the opening-range low (wider)	stop at the range midpoint
S03	Stop rule	Stop 1.5 x ATR(14) of the 5-minute bars under the entry	stop at the range midpoint
S04	Stop rule	Stop 1 % under the entry	stop at the range midpoint
S05	Target	Target at 1.5R	target at 1R
S06	Target	Target at 2R (trader-exit's own default)	target at 1R
S07	Entry order	Limit entry at the cap (size.py --limit cap)	market entry
S08	Opening range	30-minute opening range	15-minute range
S09	Volume check	Breakout bar needs 1.5 x the range bars' average volume	no volume check
S10	Setup	Level setup: a close above the previous session's high, stop 1 % under	opening-range breakout, midpoint stop
S11	Risk per trade	$100 max loss a trade ($300 a day)	$50 a trade ($150 a day)
S12	Entry window	No new entries after 11:00 (morning only)	entries 09:45 to 15:30
S13	Time Machine filter	Trade only when the Time Machine has target-first above stop-first	every READY ticket taken

Choices left out: 1-minute bars (the repo has no 1-minute history), the Time Machine's block bootstrap (it changes the forecast, not the trade), and the other config limits, which one trade a day never reaches.

Results at a glance

Scenario	Trades	Target / stop / 15:55	Won	Net	Avg R	Worst trade	Deepest open loss	Max drawdown	End cash
S01 Baseline	19	6 / 9 / 4	42 %	-$62.52	-0.18	-$48.23	-$48.23	-$138.80	$4,937.48
S02 Stop: range low	19	1 / 4 / 14	53 %	-$49.53	-0.08	-$44.52	-$44.82	-$205.04	$4,950.47
S03 Stop: 1.5 x ATR	19	9 / 7 / 3	47 %	-$33.78	+0.02	-$49.60	-$49.60	-$164.86	$4,966.22
S04 Stop: 1 %	19	7 / 6 / 6	58 %	+$50.74	+0.07	-$37.60	-$37.60	-$106.36	$5,050.74
S05 Target: 1.5R	19	2 / 9 / 8	42 %	-$47.90	-0.14	-$48.23	-$48.23	-$142.17	$4,952.10
S06 Target: 2R	19	0 / 10 / 9	37 %	-$119.50	-0.27	-$48.23	-$48.23	-$210.70	$4,880.50
S07 Entry: limit at cap	19	6 / 9 / 4	42 %	-$53.40	-0.17	-$48.23	-$48.23	-$137.08	$4,946.60
S08 Range: 30 minutes	17	2 / 7 / 8	41 %	-$88.32	-0.21	-$48.60	-$48.60	-$194.22	$4,911.68
S09 Volume: 1.5 x range	2	0 / 1 / 1	0 %	-$42.71	-0.80	-$23.00	-$26.80	-$42.71	$4,957.29
S10 Setup: prior-day high	17	8 / 6 / 3	53 %	+$59.42	+0.10	-$37.60	-$37.60	-$81.04	$5,059.42
S11 Risk: $100 a trade	17	5 / 9 / 3	41 %	-$102.10	-0.26	-$62.01	-$62.01	-$209.90	$4,897.90
S12 Window: until 11:00	17	6 / 9 / 2	35 %	-$74.26	-0.23	-$48.23	-$48.23	-$150.54	$4,925.74
S13 Time Machine filter	17	6 / 7 / 4	47 %	-$32.25	-0.09	-$48.23	-$48.23	-$125.93	$4,967.75

Avg R: the average result per trade in units of the risk the risk check approved for it (+1 is a full target, -1 a full stop). Drawdown: the largest fall of the account from a high to a later low.

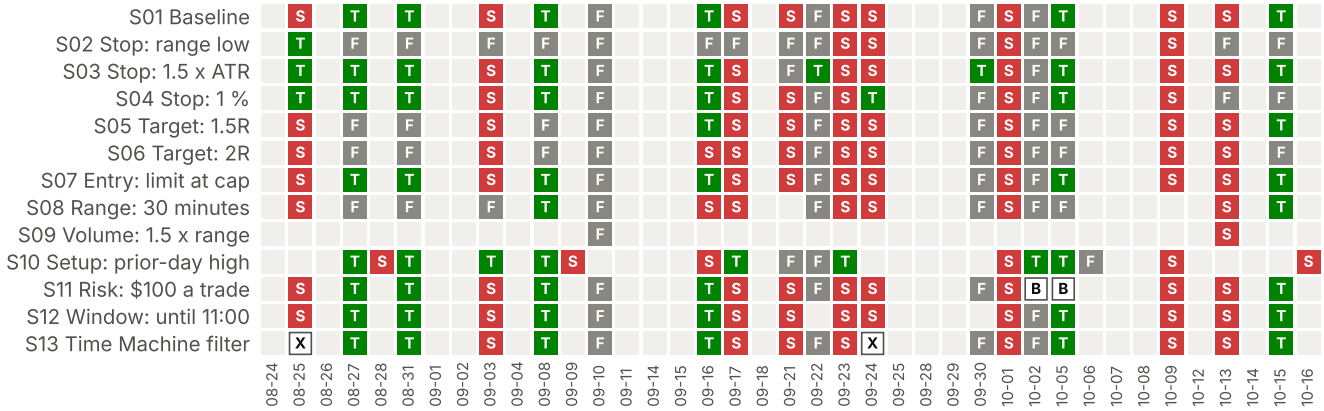


Net result over the 39 sessions (dot; blue a gain, red a loss) and its 90 % bootstrap range (bar)

The bar is the range where 90 % of 10,000 bootstrap totals fell: the scenario's own trades drawn again at random, as if the same kind of days had come in another mix. Every bar with more than a handful of trades runs from a loss to a gain, so these totals cannot rank the choices. S09 has only two trades, so its bar says little.

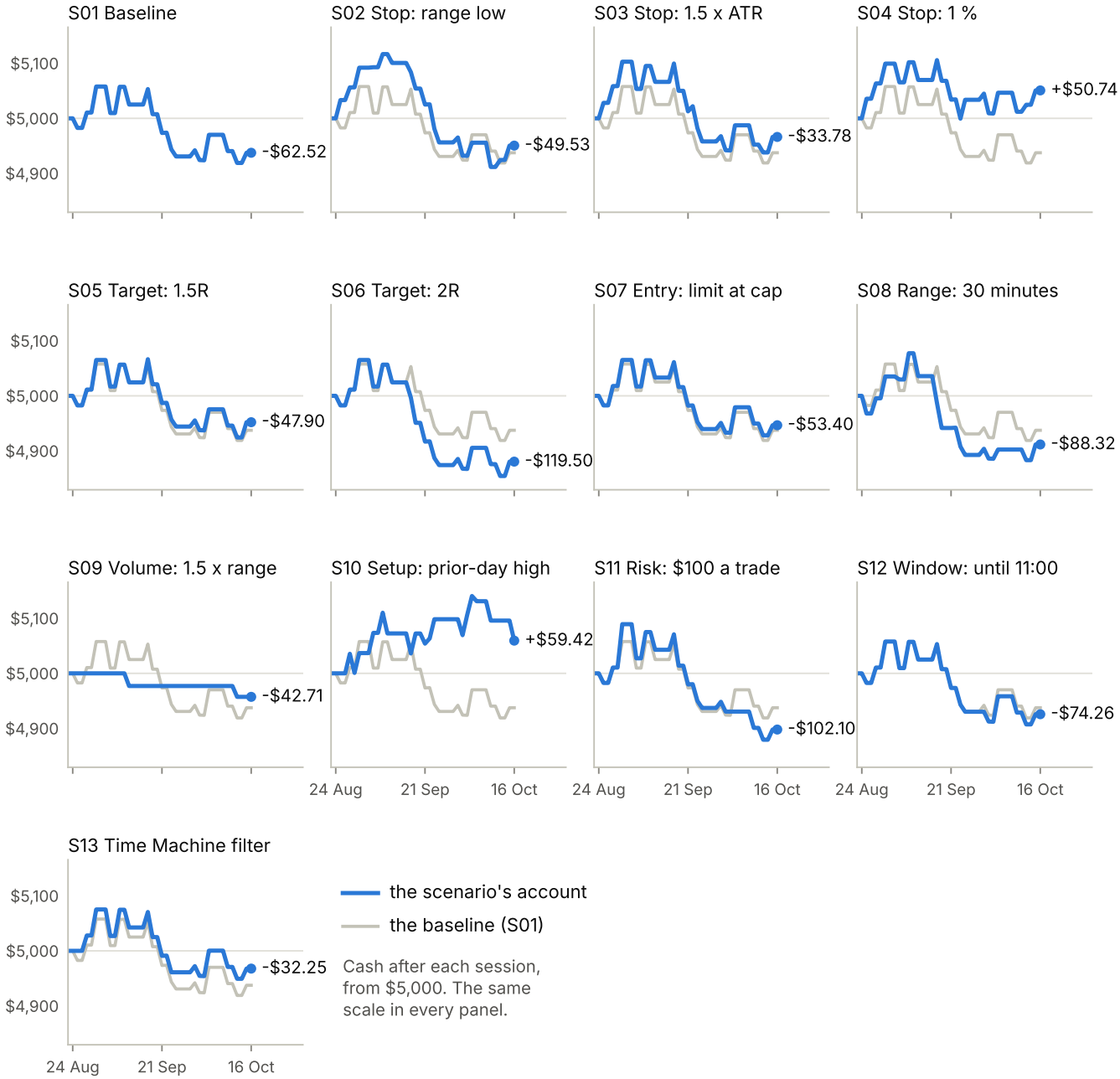
Session by session

What each scenario did on each of the 39 sessions. T target S stop F sold at the 15:55 flat-by B blocked by the risk check X skipped by the filter no signal



- The opening-range rules signal on the same 19 sessions (17 with the 30-minute range or the 11:00 cut-off), so their rows line up: a change of stop or target changes how a trade ends, not whether there is one.
- S09 and S10 trade different days. The volume check passed on only 2 sessions; the prior-day high triggered on 17, 4 of them sessions with no opening-range signal.
- No signal means the price never closed above the range high (or the level) inside the entry window: 20 of the 39 sessions for the baseline.

The account over the 39 sessions



Each step is one trade of a few tens of dollars, so a line's shape comes from a handful of sessions. S04 and S10 finish above $5,000; S06 Target: 2R finishes lowest, at $4,880.50.

What happened in each case

S01 Baseline

The README's rules: 15-minute range, stop at the range midpoint, 1R target, market entry, $50 a trade

19 trades: 6 target, 9 stop, 4 at 15:55. Net -$62.52, avg R -0.18, worst trade -$48.23, deepest open loss -$48.23.

12 of 19 orders were sized by the 75 % position cap, not the $50 max loss, so a typical trade risked about $33.52, not $50. Every other scenario is measured against this one.

S02 Stop: range low

Stop at the opening-range low (wider)

19 trades: 1 target, 4 stop, 14 at 15:55. Net -$49.53, avg R -0.08, worst trade -$44.52, deepest open loss -$44.82.

The wider stop puts the 1R target twice as far away, so most trades ran out of day: 14 of 19 were sold at the 15:55 flat-by and only 1 reached the target. Fewer shares fit the $50 limit (about 7 a trade). Against the baseline: 19 sessions came out differently, +$12.99 in all. Biggest swing: 25 Aug, a stop became a target (+$50.60).

S03 Stop: 1.5 x ATR

Stop 1.5 x ATR(14) of the 5-minute bars under the entry

19 trades: 9 target, 7 stop, 3 at 15:55. Net -$33.78, avg R +0.02, worst trade -$49.60, deepest open loss -$49.60.

An ATR stop sat near the range midpoint on most days, so most sessions ended the same way as the baseline (15 of 19). It also had the deepest open loss of any $50 scenario. Against the baseline: 19 sessions came out differently, +$28.74 in all. Biggest swing: 25 Aug, a stop became a target (+$45.50).

S04 Stop: 1 %

Stop 1 % under the entry

19 trades: 7 target, 6 stop, 6 at 15:55. Net +$50.74, avg R +0.07, worst trade -$37.60, deepest open loss -$37.60.

A 1 % stop was tighter than the midpoint on some days and wider on others. Two baseline stops became targets (25 Aug and 24 Sep). One of two scenarios that ended above $5,000. Against the baseline: 16 sessions came out differently, +$113.26 in all. Biggest swing: 25 Aug, a stop became a target (+$53.00).

S05 Target: 1.5R

Target at 1.5R

19 trades: 2 target, 9 stop, 8 at 15:55. Net -$47.90, avg R -0.14, worst trade -$48.23, deepest open loss -$48.23.

Two trades reached 1.5R and paid more than at 1R. Four of the baseline's six targets never got there and were sold at 15:55 instead: two for a little more than 1R would have paid, two for less. Against the baseline: 6 sessions came out differently, +$14.62 in all. Biggest swing: 16 Sep, a target, for a different amount (+$14.00).

S06 Target: 2R

Target at 2R (trader-exit's own default)

19 trades: 0 target, 10 stop, 9 at 15:55. Net -$119.50, avg R -0.27, worst trade -$48.23, deepest open loss -$48.23.

No trade reached 2R in these 39 sessions. The 9 that did not stop out were sold at 15:55, and 16 Sep, a baseline target, became a stop. The worst result of the 13. Against the baseline: 6 sessions came out differently, -$56.98 in all. Biggest swing: 16 Sep, a target became a stop (-$56.00).

S07 Entry: limit at cap

Limit entry at the cap (size.py --limit cap)

19 trades: 6 target, 9 stop, 4 at 15:55. Net -$53.40, avg R -0.17, worst trade -$48.23, deepest open loss -$48.23.

Every limit order filled within five minutes of the signal, at or under the plan's price, so these are the baseline's trades at slightly better fills. No loss went past the risk check's figure; on other days a limit that never fills would mean a missed trade. Against the baseline: 6 sessions came out differently, +$9.12 in all. Biggest swing: 27 Aug, a target, for a different amount (+$7.40).

S08 Range: 30 minutes

30-minute opening range

17 trades: 2 target, 7 stop, 8 at 15:55. Net -$88.32, avg R -0.21, worst trade -$48.60, deepest open loss -$48.60.

Waiting 30 minutes for the range made it wider and the entries later; two sessions never triggered. 16 Sep turned from a target into a stop. Against the baseline: 13 sessions came out differently, -$25.80 in all. Biggest swing: 16 Sep, a target became a stop (-$76.70).

S09 Volume: 1.5 x range

Breakout bar needs 1.5 x the range bars' average volume

2 trades: 0 target, 1 stop, 1 at 15:55. Net -$42.71, avg R -0.80, worst trade -$23.00, deepest open loss -$26.80.

Breakout bars this heavy were rare: 2 of 39 sessions. The check skipped all six of the baseline's targets along with most of its stops, and both trades it took lost. Against the baseline: 19 sessions came out differently, +$19.81 in all. Biggest swing: 3 Sep, a stop became no trade (+$48.23).

S10 Setup: prior-day high

Level setup: a close above the previous session's high, stop 1 % under

17 trades: 8 target, 6 stop, 3 at 15:55. Net +$59.42, avg R +0.10, worst trade -$37.60, deepest open loss -$37.60.

A different setup trades different days: 13 of its 17 trading sessions are baseline trading sessions. S04 uses the same 1 % stop on the opening-range setup, so compare it with S04 as well as with the baseline. The best result of the 13, and the smallest drawdown of the scenarios with more than two trades. Against the baseline: 23 sessions came out differently, +$121.94 in all. Biggest swing: 3 Sep, a stop became a target (+$84.93).

S11 Risk: $100 a trade

$100 max loss a trade ($300 a day)

17 trades: 5 target, 9 stop, 3 at 15:55. Net -$102.10, avg R -0.26, worst trade -$62.01, deepest open loss -$62.01.

The 75 % position cap set the size of every trade (9 or 10 shares), so doubling the limit only added shares where $50 had been the tighter limit. Two orders were blocked: size.py measures the cap against the $5,000 budget, check.py against the account after losses ($4,930.52). One of them (5 Oct) was a target in the baseline. Against the baseline: 5 sessions came out differently, -$39.58 in all. Biggest swing: 5 Oct, a target became a block (-$46.32).

S12 Window: until 11:00

No new entries after 11:00 (morning only)

17 trades: 6 target, 9 stop, 2 at 15:55. Net -$74.26, avg R -0.23, worst trade -$48.23, deepest open loss -$48.23.

Two signals came after 11:00 (22 Sep and 30 Sep); both were small flat-by gains in the baseline, so cutting them cost a little. Against the baseline: 2 sessions came out differently, -$11.74 in all. Biggest swing: 30 Sep, a 15:55 exit became no trade (-$11.20).

S13 Time Machine filter

Trade only when the Time Machine has target-first above stop-first

17 trades: 6 target, 7 stop, 4 at 15:55. Net -$32.25, avg R -0.09, worst trade -$48.23, deepest open loss -$48.23.

The filter skipped two trades (25 Aug and 24 Sep) where the Time Machine had stop-first ahead; both were stops in the baseline. Two skips are far too few to say the filter works. Against the baseline: 2 sessions came out differently, +$30.27 in all. Biggest swing: 25 Aug, a stop became a skip (+$17.40).

Maximum losses incurred at once

A trade's loss is not one number. Four ways to measure it, from the single worst moment to a bad stretch:

1. **Deepest open loss**: the most an open position was down at any one moment, marked at each 5-minute bar's low. This is the loss at once. It counts even when the trade later recovered.
2. **Worst trade**: the largest loss once a trade was closed.
3. **Risk beyond the check's figure**: check.py approves a max loss at the plan's entry price. A market order then fills at the next price, sometimes higher, and a stop at the planned price would lose more than the check approved. Shown as the largest such excess, and the count of closed losses that went past the check's figure.
4. **Drawdown and losing streak**: the largest fall of the account from a high to a later low, and the most losing trades in a row.

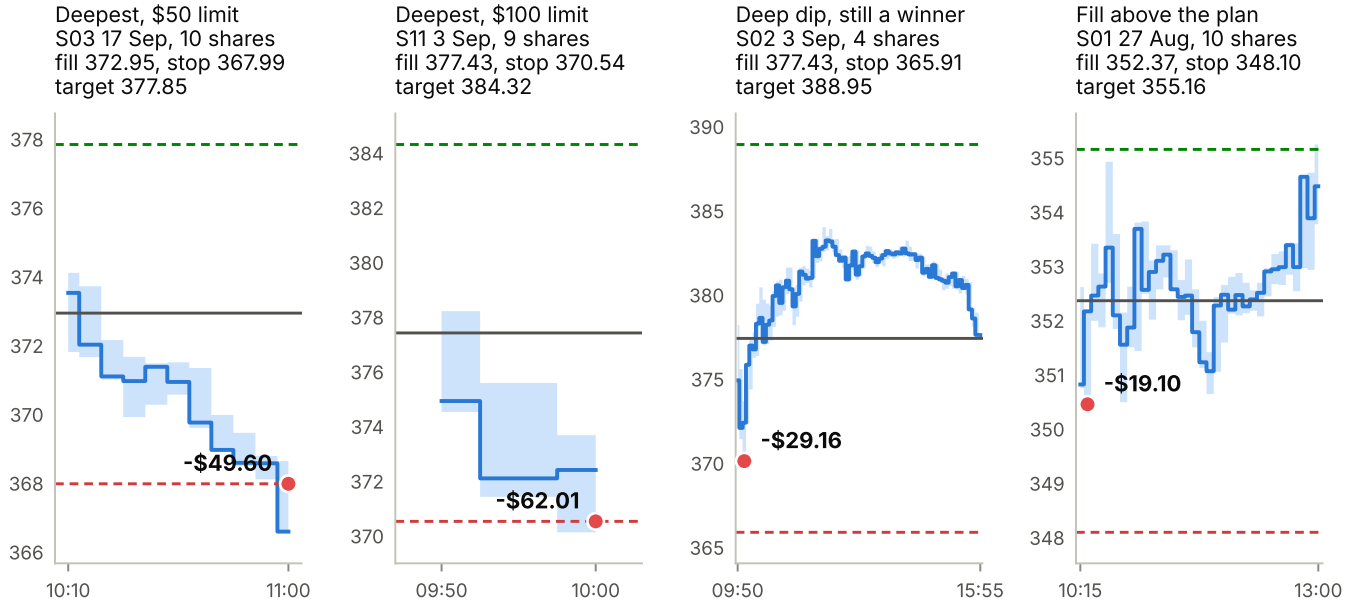
Scenario	Limit	Deepest open loss	When	Of the limit	Worst trade	Most risk beyond the check (day)	Losses past it	Max drawdown	Drawdown from, to	Losing streak
S01 Baseline	$50	-$48.23	3 Sep 10:00	96 %	-$48.23	$7.40 (27 Aug)	3	-$138.80	31 Aug to 14 Oct	2
S02 Stop: range low	$50	-$44.82	16 Sep 15:25	90 %	-$44.52	$5.92 (27 Aug)	1	-$205.04	8 Sep to 9 Oct	4
S03 Stop: 1.5 x ATR	$50	-$49.60	17 Sep 11:00	99 %	-$49.60	$7.40 (27 Aug)	2	-$164.86	31 Aug to 13 Oct	2
S04 Stop: 1 %	$50	-$37.60	17 Sep 10:45	75 %	-$37.60	$7.40 (27 Aug)	3	-$106.36	16 Sep to 23 Sep	2
S05 Target: 1.5R	$50	-$48.23	3 Sep 10:00	96 %	-$48.23	$7.40 (27 Aug)	3	-$142.17	16 Sep to 14 Oct	2
S06 Target: 2R	$50	-$48.23	3 Sep 10:00	96 %	-$48.23	$7.40 (27 Aug)	3	-$210.70	31 Aug to 14 Oct	4
S07 Entry: limit at cap	$50	-$48.23	3 Sep 10:00	96 %	-$48.23	none	0	-$137.08	31 Aug to 13 Oct	2
S08 Range: 30 minutes	$50	-$48.60	16 Sep 14:35	97 %	-$48.60	$0.42 (31 Aug)	2	-$194.22	8 Sep to 13 Oct	3
S09 Volume: 1.5 x range	$50	-$26.80	10 Sep 15:50	54 %	-$23.00	$0.36 (13 Oct)	1	-$42.71	start to 16 Oct	2
S10 Setup: prior-day high	$50	-$37.60	9 Sep 10:10	75 %	-$37.60	$0.80 (3 Sep)	2	-$81.04	5 Oct to 16 Oct	3
S11 Risk: $100 a trade	$100	-$62.01	3 Sep 10:00	62 %	-$62.01	$7.40 (27 Aug)	3	-$209.90	31 Aug to 14 Oct	3
S12 Window: until 11:00	$50	-$48.23	3 Sep 10:00	96 %	-$48.23	$7.40 (27 Aug)	3	-$150.54	31 Aug to 13 Oct	6
S13 Time Machine filter	$50	-$48.23	3 Sep 10:00	96 %	-$48.23	$7.40 (27 Aug)	3	-$125.93	31 Aug to 14 Oct	2



What those numbers say

- **No loss went past $50.** In the twelve $50 scenarios the deepest any position went was -$49.60, 99 % of the limit: the sizing works from the stop, and no bar in these sessions opened below a stop, so no stop filled worse than its price. One position did stand to lose more: on 27 Aug in S02 (Stop: range low), the market order filled 74 cents above the plan's price, so its 8 shares, sized to $49.44 at the plan, risked $55.36 at the stop. It never got there (sold at 15:55, +$23.04).
- **The deepest moment was usually the stop-out itself.** Because the stop is the floor of the trade, the worst open loss and the worst trade are the same trade in 12 of 13 scenarios. Winners dipped too: the deepest dip of a trade that still ended in profit was -$29.16.
- **Market fills carried a little more risk than the check approved.** The market order fills at the next price, which in the baseline averaged 3.7 cents a share above the plan's entry. That raised the risk at the stop by up to $7.40 on one trade (S01, 27 Aug: a fill 74 cents above the plan). Closed losses went past the check's figure by 20 to 36 cents. The limit-at-cap entry (S07) had none: it never fills above the plan's price.
- **Raising the limit to $100 raised the worst moment.** S11's deepest open loss was -$62.01, more than any $50 scenario, although the position cap kept the size at 9 or 10 shares.
- **Losses come in runs.** The drawdowns ($42.71 to $210.70) are two to four times one trade's limit, spread over days. The $150 daily loss limit counts one day at a time, so it never stops a run like this: with one trade a day it was never reached.

Four of those moments, bar by bar



The blue line is each bar's close and the pale band its low-to-high range, from the fill to the exit. The red dot is the deepest moment. Dashed lines: the target (green) and the stop (red); the solid gray line is the fill.

- **Deepest, $50 limit** (S03 Stop: 1.5 x ATR, 17 Sep): the deepest open loss of the twelve $50 scenarios, -$49.60, at the stop-out
- **Deepest, $100 limit** (S11 Risk: $100 a trade, 3 Sep): -$62.01 at once with the doubled limit
- **Deep dip, still a winner** (S02 Stop: range low, 3 Sep): down $29.16 at one point before it ended +$0.80
- **Fill above the plan** (S01 Baseline, 27 Aug): filled 0.74 over the plan's entry, so the stop risked $42.70 against the check's $35.30; it hit the target

The Time Machine's forecasts against what happened

For every trade, the Time Machine replayed the rest of the session 10,000 times from TSLA's own earlier sessions and gave the odds of the target first, the stop first, or neither by the close. Averaged over each scenario's trades, next to what the trades did:

Scenario	Forecast: target first	Stop first	Target share of the two	Trades that hit target, of those that hit one	Actual target share
S01 Baseline	45 %	38 %	54 %	6 of 15	40 %
S02 Stop: range low	28 %	21 %	57 %	1 of 5	20 %
S03 Stop: 1.5 x ATR	48 %	40 %	55 %	9 of 16	56 %
S04 Stop: 1 %	49 %	40 %	55 %	7 of 13	54 %
S05 Target: 1.5R	30 %	42 %	41 %	2 of 11	18 %
S06 Target: 2R	20 %	44 %	31 %	0 of 10	0 %
S07 Entry: limit at cap	45 %	38 %	54 %	6 of 15	40 %
S08 Range: 30 minutes	42 %	33 %	56 %	2 of 9	22 %
S09 Volume: 1.5 x range	50 %	45 %	53 %	0 of 1	0 %
S10 Setup: prior-day high	55 %	41 %	57 %	8 of 14	57 %
S11 Risk: $100 a trade	46 %	39 %	54 %	5 of 14	36 %
S12 Window: until 11:00	47 %	39 %	55 %	6 of 15	40 %
S13 Time Machine filter	44 %	36 %	55 %	6 of 13	46 %

- For the 1R scenarios with five or more trades that hit a side (all but S09), the forecast's target share sat between 54 % and 57 %; what happened was 20 % to 57 %. Twenty trades or fewer per scenario cannot confirm or refute a forecast that close to a coin flip.
- The forecast moved with the rule: a farther target lowered the forecast odds of reaching it, from 45 % in the baseline to 28 % (S02, range-low stop), 30 % (1.5R) and 20 % (2R), and the targets actually reached fell with it, from 6 to 1, 2 and 0.
- As a filter (S13), the Time Machine had stop-first ahead on only two of the 19 signals, and both of those trades were stops in the baseline. It is one way to use the forecast, not proof that it picks trades.

The Time Machine draws its replays only from sessions before the trade (its pool grew from 21 to 57 sessions over the run), so it never saw the day it was forecasting.

Things the runs turned up in the tools

1. **size.py and check.py measure the 75 % position cap differently.** size.py takes 75 % of the $5,000 budget ($3,750); check.py takes 75 % of the account, which after losses was $4,930.52 ($3,697.89). In S11, size.py proposed 10 shares twice (about $3,713 and $3,741), and check.py blocked both (2 Oct and 5 Oct). In real use you would see a BLOCK on an order the budget tool had just written. Not changed: tell me if you want the two to agree.
2. **A market fill above the plan's price is risk the check did not approve.** check.py approves the max loss at the plan's entry (or a fresh price); the fill can be higher. Here it added up to $7.40 of risk on one trade, and once it took a trade's risk past the $50 limit ($55.36 at the stop, S02 on 27 Aug). A fast market makes the gap bigger. `size.py --limit cap` (S07) bounds it, and on these days every such limit filled within five minutes.
3. **An EXPIRED entry watch leaves yesterday's entry.json in place.** The watch writes the file only on a trigger, so exit.py plan goes on to plan the old entry with only a note ("This plan is for 2026-08-03, not today"). size.py then refuses the old plan ("make a new plan with trader-exit"), so no order can come of it; it costs a confusing step. The runs here clear the folder each morning. Not changed.
4. **size.py's report names "trader/portfolio.json"** as the source of the cash even when `--portfolio` or TRADER_HOME points somewhere else. Cosmetic. Not changed.
5. **Fixed: the tests and TRADER_HOME.** With TRADER_HOME set, 12 tests failed and the tests wrote `entry.json`, `plan.json` and `timemachine.json` into that folder, which would have overwritten a real trader folder. `tests/_load.py` now clears it for the test run; 384 tests pass either way.

Method and limits

- **One stock, 39 sessions.** TSLA from 24 Aug to 16 Oct 2026; the scanner was not part of the runs, because the repo has a market-wide scan for only one morning (19 Oct). The days and the stock are the README example's.
- **One trade a day at most.** The opening-range rule signals once a session, so the daily limits (3 trades, $150, 10 orders) were never reached and concurrent positions never happened.
- **Simulated fills.** From 5-minute bars: no fees, no spread, no partial fills, and the stop is assumed first when a bar reaches both legs (it never happened here). Real fills, especially market orders and stops in a fast market, can be worse.
- **Practice clock.** Every check.py and ticket.py run was a practice run (pinned clock); their verdicts would never count for a real order, by design.
- **Reproducible.** `python -B paper-trading-scenarios/run_scenarios.py` reruns all 13 in about 3 minutes with the same numbers; `make_report.py` rebuilds this PDF. Every command and its full output is in each scenario's `run.log`, every session in `trades.csv`, and every tool file of every trading day under `days/`.

Education and tooling, not financial advice.